

Module 3 Assignment — Infrastructure & Containerization

Spam-Detection API: Docker & Kubernetes

Q1 Single-Stage vs. Multi-Stage Docker

1. The naive Dockerfile (FROM python:3.11) builds and runs correctly, serving /healthz and /predict correctly. Image size: **240 MB**.
2. The multi-stage Dockerfile (venv built in a python:3.11-slim builder stage, copied into a python:3.11-slim runtime stage) also runs correctly. Image size: **163 MB**, a **32.1%** reduction.

spam-api:multistage	50fc3572ea3c	719MB	163MB
spam-api:naive	f2b97b9ef197	866MB	240MB

3. Why it is smaller: The naive image uses the full python:3.11 base and keeps the pip download/build cache and wheel-build artifacts (from compiling scikit-learn/numpy/scipy) in its final layer. The multi-stage build isolates all of this inside the discarded builder stage, copying only the finished /opt/venv environment and application code into a fresh slim runtime stage.

Q2 Multi-Container Orchestration with Docker Compose

1. /predict checks Redis for the exact input text; on a miss it computes the prediction, stores it with a TTL, and returns it; on a hit it returns the cached label directly.
2. docker-compose.yml defines two services: api (built from the Question 1 multi-stage Dockerfile) and cache (redis:7-alpine); the API reaches Redis via the service name cache as its hostname.

```
(spam-api) sreekar@sreekar-ASUS-Zenbook-14-UX3405CA-UX340:~/Documents/Academics/Sem-5/AI_Ops/Assignments/DA3408-Assignment-2/Question-2$ time curl -X POST http://localhost:8080/predict -H "Content-Type: application/json" -d '{"text":"WIN a FREE iPhone now!","kt":"WIN a FREE iPhone now!"}'
{"label":"spam"}
real    0m0.055s
user    0m0.005s
sys     0m0.009s
(spam-api) sreekar@sreekar-ASUS-Zenbook-14-UX3405CA-UX340:~/Documents/Academics/Sem-5/AI_Ops/Assignments/DA3408-Assignment-2/Question-2$ time curl -X POST http://localhost:8080/predict -H "Content-Type: application/json" -d '{"text":"WIN a FREE iPhone now!","kt":"WIN a FREE iPhone now!"}'
{"label":"spam"}
real    0m0.021s
user    0m0.004s
sys     0m0.010s
```

3. Evidence: An identical repeated request took **0.055 s** (cache miss) on the first call and **0.021 s** (cache hit) on the second — a measurable, repeatable speedup.

4. Compose vs. Kubernetes: Compose declaratively runs and networks multiple related containers on a *single host*, replacing manual docker run/--network commands. Kubernetes instead schedules containers across a *cluster* of nodes and provides self-healing and rolling updates, which are outside Compose's scope.

Q3 Kubernetes Indexed Job: Parallel Data Validation

1. With 4 allocatable CPUs (2 nodes × 2 CPUs) and a **250m** CPU request per pod, **parallelism: 8** was chosen so all 8 shards validate in one concurrent wave ($8 \times 250\text{m} = 2000\text{m} \leq 4000\text{m}$); the Downward API exposes POD_NAME and NODE_NAME to each pod.
2. kubectl get pods -o wide confirmed all 8 pods reached Running within the same 1–2 s window, spread across both cluster nodes, before completing.

3. Results collection: Results were collected via the Kubernetes API (reading each pod's logs) rather than a shared volume, because minikube's default storage provisioner creates node-local hostPath volumes that are not visible across nodes; the API server retrieves logs directly from whichever node a pod ran on regardless of placement.

4. **At 6 CPUs (3 nodes × 2 CPUs):** Parallelism would remain 8, since completions is fixed by the number of shards and the original request (2000m) already fit well within the 4000m budget — the extra CPU headroom does not unlock further concurrency.

Q4 Kubernetes Deployments: Self-Healing & Rolling Updates

1. A Deployment (2 replicas, CPU/memory requests and limits, a `readinessProbe` against `/healthz`) and a Service were applied successfully.

```
(spam-api) sreekar@sreekar-ASUS-Zenbook-14-UX3405CA-UX340:~/Documents/Academics/Sem-5/AI_Ops/Assignments/DA3408-Assignment-2/Question-4$ kubectl get pods -o wide
POD_TO_DELETE=$(kubectl get pods -l app=spam-api -o jsonpath='{.items[0].metadata.name}')
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE READINESS GATES
spam-api-7856558f7f-7bzsm 1/1 Running 0 62s 10.244.0.3 minikube <none> <none>
spam-api-7856558f7f-q9499 1/1 Running 0 62s 10.244.0.4 minikube <none> <none>
(spam-api) sreekar@sreekar-ASUS-Zenbook-14-UX3405CA-UX340:~/Documents/Academics/Sem-5/AI_Ops/Assignments/DA3408-Assignment-2/Question-4$ kubectl delete pod "$POD_TO_DELETE"
pod "spam-api-7856558f7f-7bzsm" deleted from default namespace
(spam-api) sreekar@sreekar-ASUS-Zenbook-14-UX3405CA-UX340:~/Documents/Academics/Sem-5/AI_Ops/Assignments/DA3408-Assignment-2/Question-4$ kubectl get pods -o wide
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE READINESS GATES
spam-api-7856558f7f-h4wxg 0/1 Running 0 6s 10.244.0.5 minikube <none> <none>
spam-api-7856558f7f-q9499 1/1 Running 0 78s 10.244.0.4 minikube <none> <none>
(spam-api) sreekar@sreekar-ASUS-Zenbook-14-UX3405CA-UX340:~/Documents/Academics/Sem-5/AI_Ops/Assignments/DA3408-Assignment-2/Question-4$ kubectl get pods -o wide
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE READINESS GATES
spam-api-7856558f7f-h4wxg 1/1 Running 0 9s 10.244.0.5 minikube <none> <none>
spam-api-7856558f7f-q9499 1/1 Running 0 81s 10.244.0.4 minikube <none> <none>
```

2. **Self-healing:** After manually deleting a pod, a replacement appeared within seconds and reached 1/1 Running, restoring the desired replica count. The **ReplicaSet controller** continuously reconciles the actual pod count against the desired replicas: 2 and schedules a new pod as soon as it detects a mismatch.

```
(spam-api) sreekar@sreekar-ASUS-Zenbook-14-UX3405CA-UX340:~/Documents/Academics/Sem-5/AI_Ops/Assignments/DA3408-Assignment-2/Question-4$ kubectl set image deployment/spam-api spam-api=spam-api:v2
deployment.apps/spam-api image updated
(spam-api) sreekar@sreekar-ASUS-Zenbook-14-UX3405CA-UX340:~/Documents/Academics/Sem-5/AI_Ops/Assignments/DA3408-Assignment-2/Question-4$ kubectl rollout status deployment/spam-api
Waiting for deployment "spam-api" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "spam-api" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "spam-api" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "spam-api" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "spam-api" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "spam-api" rollout to finish: 1 old replicas are pending termination...
deployment "spam-api" successfully rolled out
(spam-api) sreekar@sreekar-ASUS-Zenbook-14-UX3405CA-UX340:~/Documents/Academics/Sem-5/AI_Ops/Assignments/DA3408-Assignment-2/Question-4$ kubectl get pods -o wide
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE READINESS GATES
spam-api-677c68cfd-q2drg 1/1 Running 0 9s 10.244.0.7 minikube <none> <none>
spam-api-677c68cfd-wslhj 1/1 Running 0 21s 10.244.0.6 minikube <none> <none>
spam-api-7856558f7f-h4wxg 0/1 Completed 0 3m1s 10.244.0.5 minikube <none> <none>
(spam-api) sreekar@sreekar-ASUS-Zenbook-14-UX3405CA-UX340:~/Documents/Academics/Sem-5/AI_Ops/Assignments/DA3408-Assignment-2/Question-4$ kubectl get pods -o wide
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE READINESS GATES
spam-api-677c68cfd-q2drg 1/1 Running 0 12s 10.244.0.7 minikube <none> <none>
spam-api-677c68cfd-wslhj 1/1 Running 0 24s 10.244.0.6 minikube <none> <none>
(spam-api) sreekar@sreekar-ASUS-Zenbook-14-UX3405CA-UX340:~/Documents/Academics/Sem-5/AI_Ops/Assignments/DA3408-Assignment-2/Question-4$ kubectl get rs
kubectl: command not found
(spam-api) sreekar@sreekar-ASUS-Zenbook-14-UX3405CA-UX340:~/Documents/Academics/Sem-5/AI_Ops/Assignments/DA3408-Assignment-2/Question-4$ kubectl get rs
NAME DESIRED CURRENT READY AGE
spam-api-677c68cfd 2 2 2 39s
spam-api-7856558f7f 0 0 0 4m31s
(spam-api) sreekar@sreekar-ASUS-Zenbook-14-UX3405CA-UX340:~/Documents/Academics/Sem-5/AI_Ops/Assignments/DA3408-Assignment-2/Question-4$ kubectl rollout history deployment/spam-api
deployment.apps/spam-api
REVISION CHANGE-CAUSE
1 <none>
2 <none>
(spam-api) sreekar@sreekar-ASUS-Zenbook-14-UX3405CA-UX340:~/Documents/Academics/Sem-5/AI_Ops/Assignments/DA3408-Assignment-2/Question-4$ curl "$API_URL/healthz"
{"status":"ok","version":"v2"}(spam-api) sreekar@sreekar-ASUS-Zenbook-14-UX3405CA-UX340:~/Documents/Academics/Sem-5/AI_Ops/Assignments/DA3408-Assignment-2/Question-4$ curl "$API_URL/healthz"; echo
{"status":"ok","version":"v2"}
```

3. **Rolling update:** `kubectl set image` to a new tag triggered a new ReplicaSet scaling up while the old one scaled down; rollout status reported success and `/healthz` returned the updated version string with zero downtime. As a Job, the same change would require manually deleting and recreating it, causing downtime.

4. **Deployment vs. Job:** A Deployment suits long-running, always-on serving workloads that need continuous availability and safe updates, while a Job suits run-to-completion batch workloads (Question 3) that end once their unit of work finishes and should not be restarted.